

Resolution

The resolution of a bitmapped digital image is measured in pixels or dots per inch (DPI). Expanding a bitmapped image causes the pixels to become visible. As a result the picture looks blocky or pixelated. Vector images don't have this issue, since the maths functions they contain can easily be used to generate larger versions. The bit depth of a colour picture defines how many bits are used to store the binary value for the colour of a pixel.

▽ Dots per inch

An image's DPI value defines how detailed it is. A higher value denotes a more detailed image that can be made larger, with no clarity lost.



72 DPI



150 DPI



300 DPI

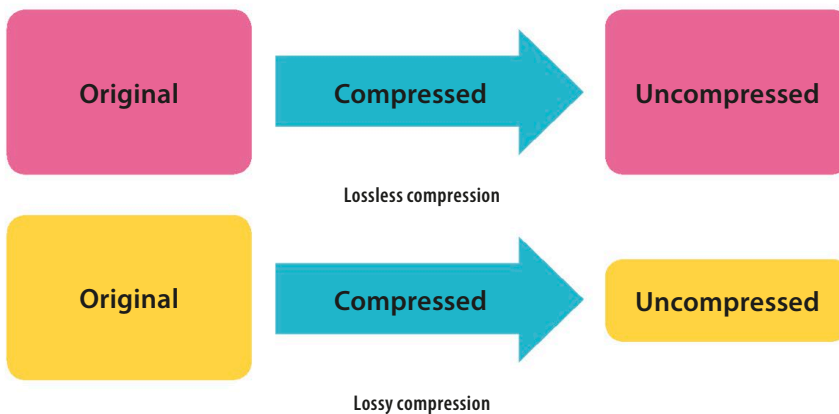
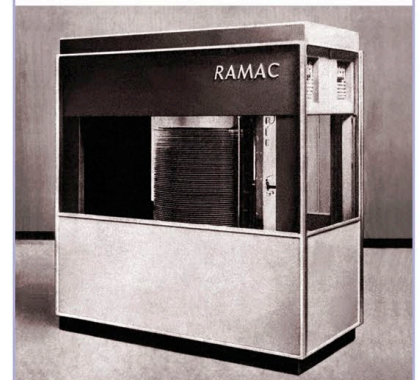
Why are image files so large?

As cameras get better at capturing detail, the resolution of pictures increases and so does the amount of data needed to store these details. RGB images use at least 24 bits for each pixel to store colour data, increasing file sizes even further. Files from professional digital cameras can be around 40 MB, and a high-resolution jpeg can be around 12 MB.

IN DEPTH

A history of data

Developed in 1956, the IBM RAMAC 350 was the first computer to have a magnetic disk drive similar to those used today. It weighed 1 tonne (1 ton) and had 50 disks that stored a total of 5 MB of data. That's around the file size of one high-resolution jpeg photo. Today, people can carry gigabytes of data in their pockets on their cell phones.



△ Compression

It's possible to compress files to make them easier to share or store. Compression can be either lossless or lossy. Lossless means the picture quality remains the same, with no information lost. Lossy is the opposite: some information is lost to make the file size smaller, which affects the picture quality.